Alternative Assessment Tool (AAT) Report on

"Design and Verification of Combinational and Sequential Digital Modules using Verilog HDL"

Submitted in partial fulfilment of the requirements of the Alternative Assessment Tool (AAT) for the course

ADVANCED DIGITAL AND LOGIC DESIGN (ADLD)

[Course Code: 23ECSPE1AD]

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Academic Year

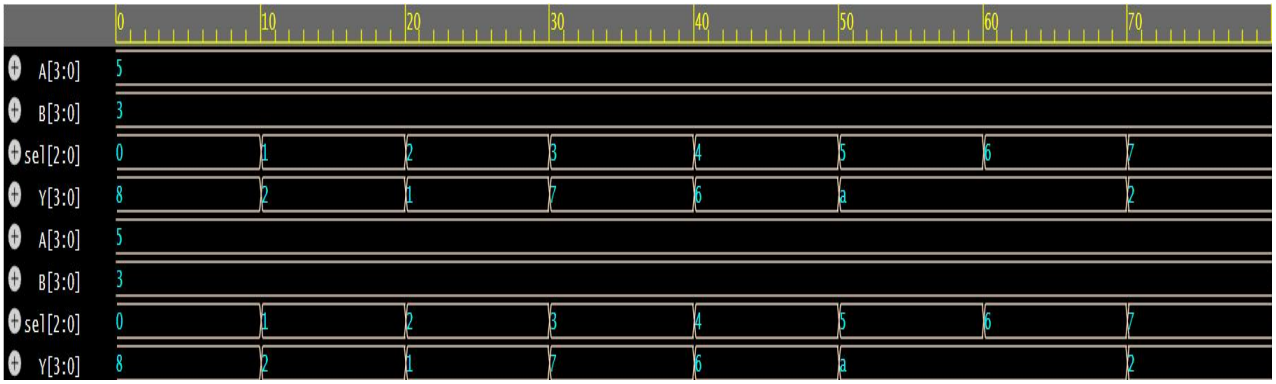
2026 – 2027

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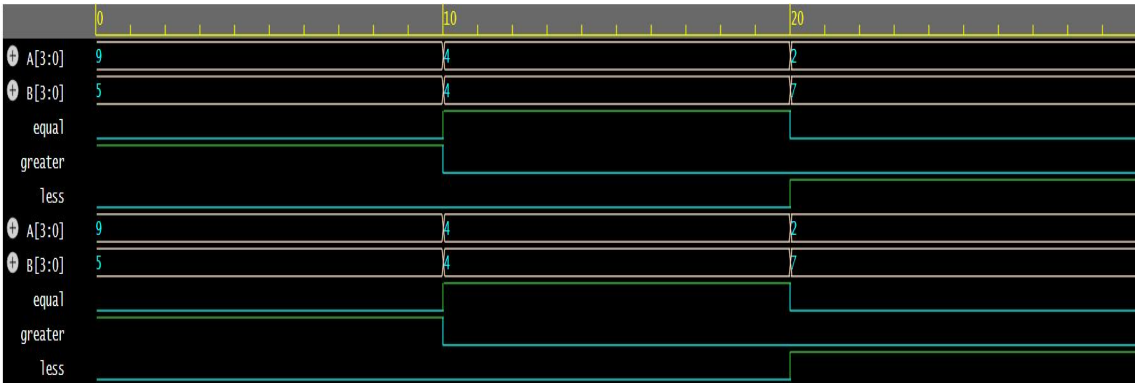
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SIMULATION

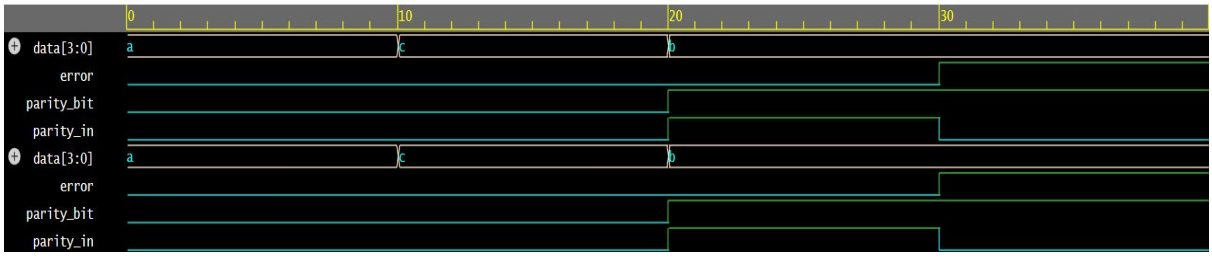
1. ARITHMETIC LOGIC UNIT (ALU)



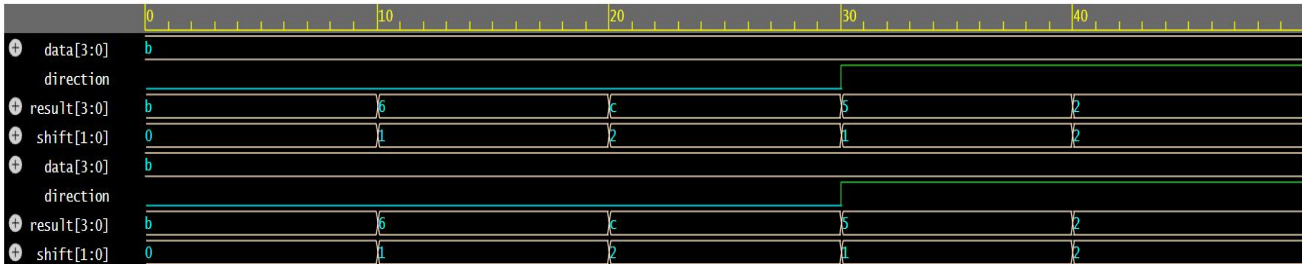
2. FOUR-BIT COMPARATOR



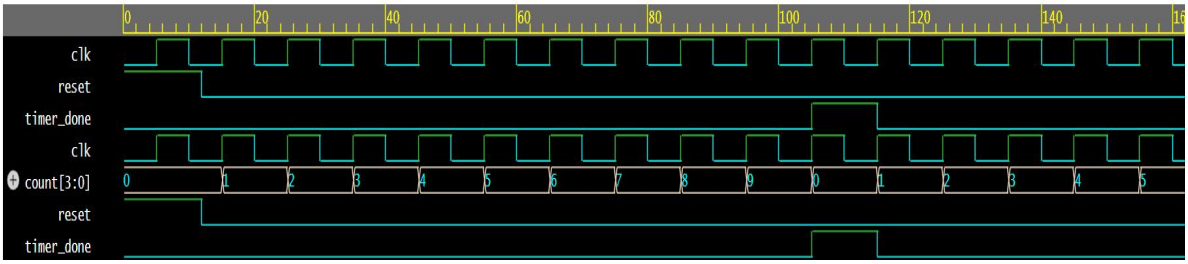
3. FOUR-BIT PARITY GENERATOR AND CHECKER



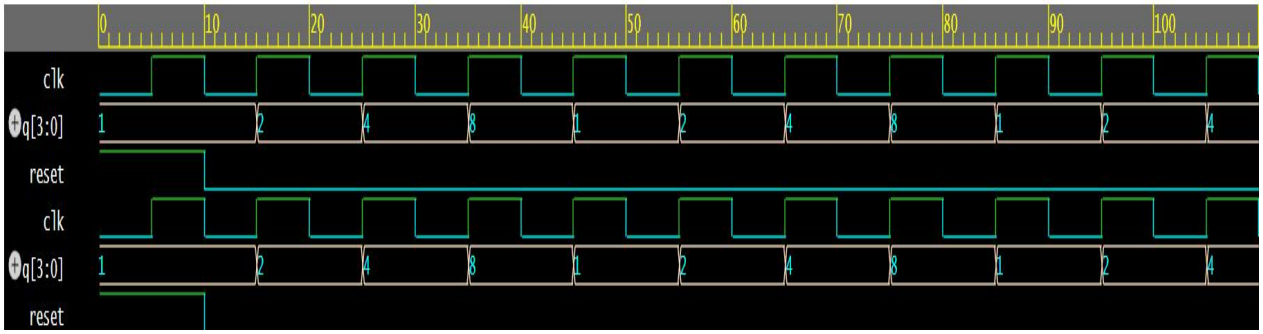
4. FOUR-BIT BARREL SHIFTER



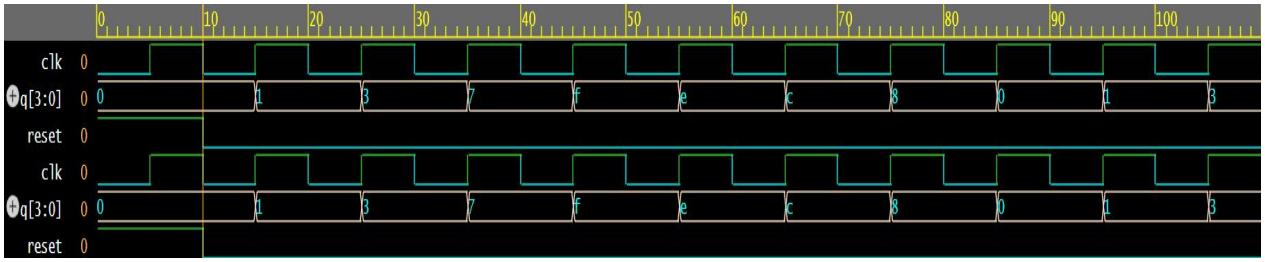
5. TIMER MODULE



6. FOUR-BIT RING COUNTER

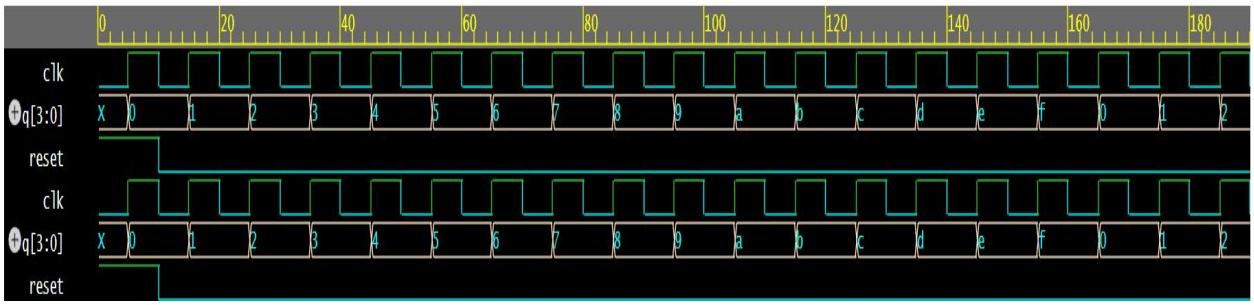


7. Johnson Counter

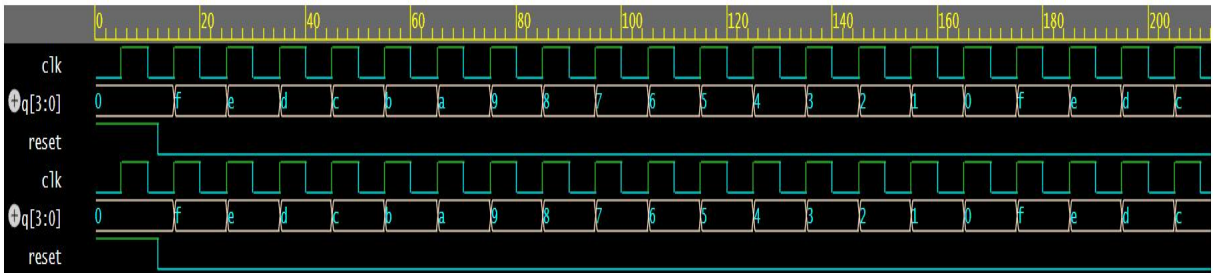


8. Synchronous and Asynchronous Counters

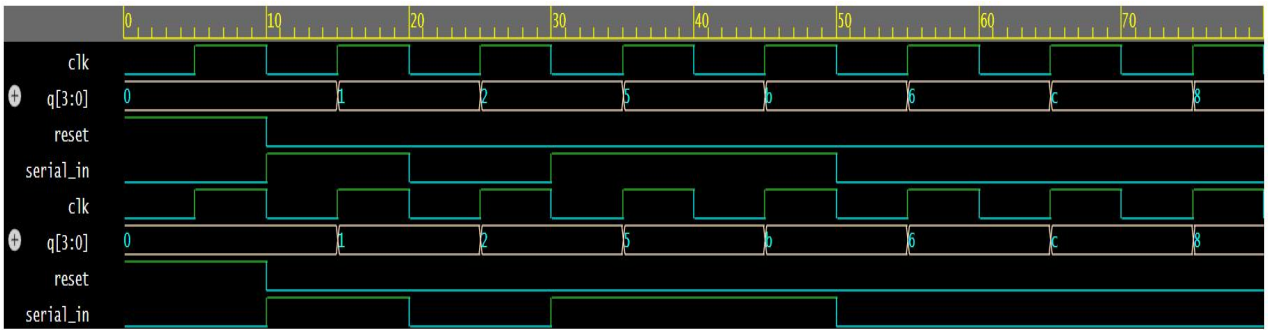
A. Synchronous 4-Bit Up-Counter



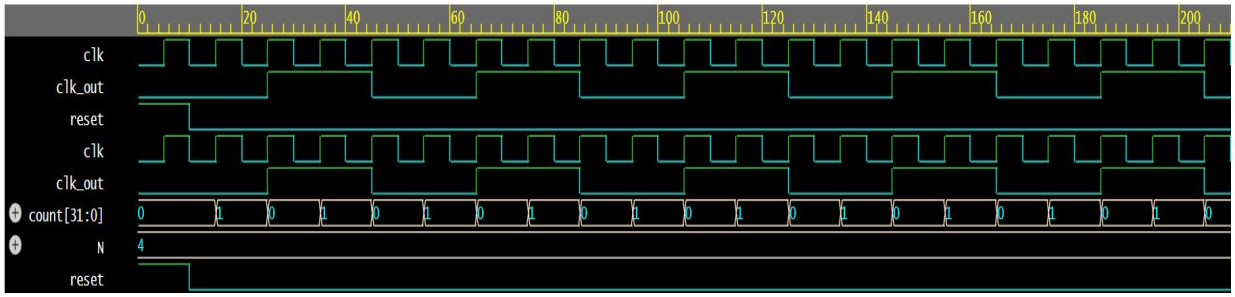
B. Asynchronous 4-Bit Ripple Up-Counter



9. Four-Bit Shift Register

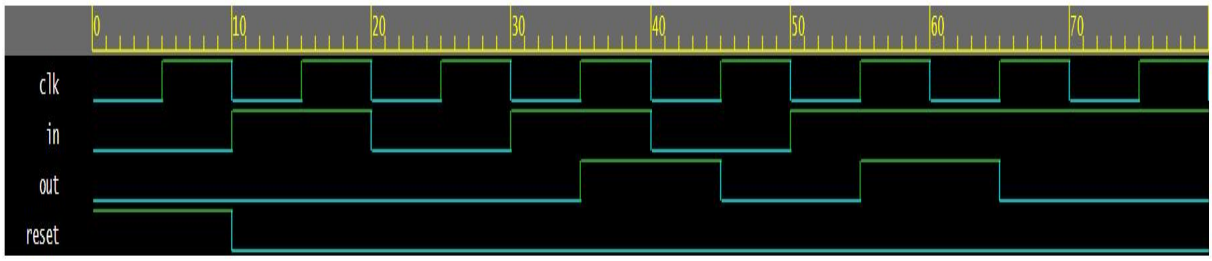


10. Divide-by-N Counter

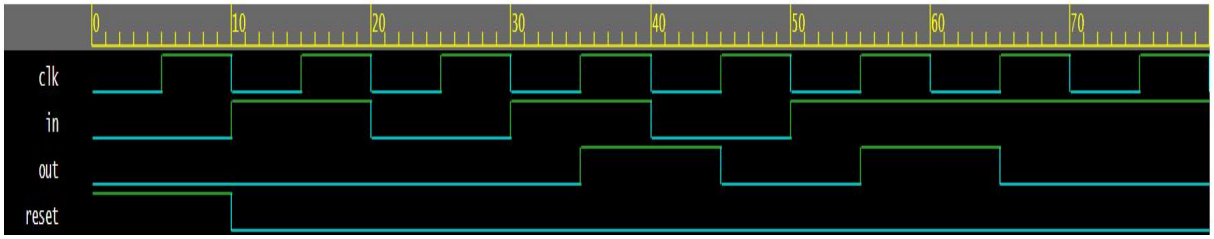


11. Mealy and Moore Finite State Machines (FSMs)

A. Mealy FSM — Sequence Detector for 101



B. Moore FSM — Sequence Detector for 101



12. Random Sequence Generator (4-Bit LFSR)

